



Session 4 – Thinking London Spatially (Spatial Query with SpatialLite Basics)

Objectives: To gain familiarity in spatial query with SQL through SpatialLite. London Data are used in various cases across 33 boroughs.

Task 1: Building a Spatial Database

Create a new SQLite Database, Name it as “test.sqlite”

Load Shapefile (“london_boroughs.shp”, “london_io.shp” and “UndergroundStations.shp”),

Load CSV/TXT (“london_pophh.csv”)

Task 2: Basic SQL Queries

Run basic SQL queries on London Population, London Households and London Underground Stations.

Get familiar with SQL commands of Select, From, Where, Order, Group.

Task 3: Basic Spatial SQL Queries

Run basic spatial SQL queries on London population and households.

Try to create the Point geometry from a CSV table, select the geometries and calculate the area of geometry.

Task 4: Create new tables for London population

Create a simple table, Create a mappable table with geometry column, Create a table with constraints.

Task 5: Edit the table of London population

Try Insert, Update and Delete rows in a table.

Task 6: Join tables of different London data

Join 2 tables, Join 3 tables, Join tables with more conditions, Left join.

Task 7: Create a View of London population

Create a view, Query a view, Join a table with a view.

Disconnect the current SQLite Database